

Efficient Joint Beam Pattern and Power Allocation in Beam Hopping LEO Satellite Systems: A Multi-Agent Actor-Critic Approach

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Abstract—Multi-beam low Earth orbit (MB-LEO) satellites are crucial for satellite Internet access, as they simultaneously transmit multiple spot beams to serve numerous ground users. The creation of dispersed beam patterns to avoid interference and the efficient allocation of limited onboard resources significantly impact the quality of communication services. Traditional pre-defined periodic beam hopping (BH) methods struggle with unevenly distributed and time-varying traffic requests. Existing BH resource allocation schemes using reinforcement learning face challenges in avoiding inter-beam interference through location selection and are constrained by action space, complicating model training as user and beam numbers increase. This paper presents a comprehensive framework and its mathematical formulation for efficient BH resource allocation in MB-LEO satellite systems using a multi-agent actor-critic method. To address the multi-objective, multi-dimensional optimization problem, the resource allocation process is divided into two stages. First, the optimal beam pattern is determined to maximize channel capacity and system throughput. Then, a power allocation (PA) agent dynamically distributes power to each beam based on traffic demand, optimizing resource utilization. During the training phase, a dynamic adjustment mechanism guides BH agents to balance throughput maximization and packet loss minimization by adjusting the penalty for dropped traffic requests. Experimental results demonstrate the effectiveness of our method in avoiding inter-beam interference, improving system throughput, reducing latency, and allocating resources based on demand.

Index Terms—Multi-beam satellite, beam hopping, power allocation, multi-agent reinforcement learning

I. INTRODUCTION

WITH the integration of air-space-ground communications, multi-beam Low Earth Orbit satellites have gained increasing attention on the satellite internet. They serve as relays between high-altitude satellites and ground users in satellite internet systems such as Starlink and OneWeb [1],

[2]. The onboard phased array antennas generate multiple independent high-gain beams provided with distributed power and bandwidth [3]–[6]. The Multi-beam low Earth orbit (MB-LEO) satellites employ the beam hopping (BH) technique, where spot beams provide services to different users in a time-division multiplexed manner [7]–[10]. Benefiting from flexible beam scheduling and low orbit, MB-LEO satellites offer advantages such as high capacity, wide coverage, low latency, and seamless communication [8], [11], [12]. However, MB-LEO satellites are severely constrained by onboard resources including power, bandwidth, and computing ability. It is essential to allocate resources efficiently to deliver high-quality and reliable communication services.

Due to the time-varying and non-uniformly geographically distributed traffic demands, as well as the wide range of possible beam illumination patterns, obtaining the optimal resource allocation scheme is a challenging task [13]–[15]. Achieving BH under the constraints of limited satellite resources, while ensuring communication capability that matches the traffic demands, reducing interference in adjacent illuminating areas, and minimizing latency, has emerged as a current research hotspot.

A. Related works

To achieve optimal resource allocation and reduce interference, researchers have explored various approaches. Aravanis *et al.* [16] proposed a two-stage multi-objective optimization approach, which obtained a set of Pareto optimal solutions by genetic algorithm and simulated annealing algorithm. This approach effectively balanced the trade-offs between different objectives and achieved good performance in terms of system throughput, power consumption, and received SINR. Han *et al.* [17] considered the Quality of Services (QoS) requirements of different beams, and proposed a novel slot allocation scheme to distribute the available slots in a way that ensures fairness and optimizes resource utilization. Feng *et al.* [18] introduced an efficient resource allocation mechanism based on a greedy algorithm for the BH system. The mechanism allocated beam, bandwidth, and power resources based on factors such as beam traffic demand, signal quality, and interference levels. However, considering the dynamic environment of the LEO satellite, the aforementioned methods were based on heuristic algorithms with high computational complexity, which require extensive computations in each decision-making process, and

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failed to meet the low-latency and real-time decision-making requirements of the MB-LEO satellite communication system.

Leveraging the promising performance of deep reinforcement learning (DRL) in complex resource allocation and the exceptional capability of obtaining fast and feasible decision results through forward inference [19], researchers have started applying DRL to satellite resource allocation. By modeling BH as a Markov decision process, they utilize well-trained agents to generate beam patterns in each time slot. Hu *et al.* [20] proposed a DRL-based framework for dynamic resource allocation in multi-beam satellite systems. Traffic demands and distributions were converted to tensor tables and convolutional neural networks (CNN) were employed to extract the spatial and temporal characteristics of traffic demands. This study highlights the advantages of DRL in dynamic resource allocation, including adapting to changing environments, handling complex interactions among multiple beams, and improving system efficiency.

Considering the problem of BH, the selection of illuminating areas of beams encountered difficulties in the single-agent DRL approach due to the exponential growth of the joint action space dimension. Specifically, when there are K beams serving N ground cells, the action space expands to C_N^K , which is untrainable for multi-beam satellite agents. Currently, two decision-making approaches are introduced to address this issue: 1) Sequential decisions: Hu *et al.* [21] [22] proposed a Double-Loop Learning framework, which reset the traffic demand of the selected cell to zero after each decision. 2) Select the K beam indices with the highest Q score from Softmax layer outputs of the deep Q network (DQN) as the joint action. But these methods did not consider the potential problem of inter-beam interference.

The single agent in [23] output beam indices and coverage areas, and utilizes a pseudo-update of the status to make decisions for all beams. It reduced inter-beam interference by controlling the beam coverage areas. Du *et al.* [24] revealed the communication performance and complexity of full frequency reuse was better than partial frequency reuse in the BH system with fewer beams, and applied an on-policy algorithm Proximal Policy Optimization (PPO) to determine the beam coverage of cells and allocate power to each beam. This approach faces challenges due to the enormous action space and slow convergence.

The multi-agent DRL methods with distributed execution have gradually attracted the attention of researchers in order to reduce action space and complexity. Liao *et al.* [25] introduced multi-agent reinforcement learning (MARL) to satellite resource allocation by modeling each beam as an individual agent and utilizing the DQN algorithm for bandwidth allocation. Lin *et al.* [26] extended this work by considering the allocation of both time slot and bandwidth resources, where the illuminating area selection and bandwidth allocation of each beam were modeled as separate agents. An independent Q-learning (IQL) approach was introduced to train the agents and the MARL system achieved a Nash equilibrium after training. Agents in the proposed method share the same reward and observation: the satellite traffic buffer, and each agent stores this information and its own action to its own replay

buffer.

However, in the problem of BH in an MARL environment, relying solely on observing awaiting traffic is insufficient for the agents to effectively learn how to avoid making the same or adjacent action. The IQL-based approach neither explicitly incorporated the decision results of fellow agents into observation, nor utilized value decomposition methods to evaluate the contribution of different agents to the environment reward. While the DQN-based method is commonly employed in the studies that apply DRL for BH, it encounters difficulties in the MARL context. Specifically, one of the challenges arises during the experience replay. In both the training and execution phases, the observations of the DQN agents are the satellite buffer. But due to changes in other agents' policies, an agent may receive a different reward for the same action despite having the same observation. This inconsistency in the training process impedes convergence among the multi-agents.

Different from IQL, the Multi-agent Deep Deterministic Policy Gradient (MADDPG) approach follows the centralized training with distributed execution paradigm [27]. The critic network has a global observation including the global state and actions of all agents, and the actor network receives partial observation, which helps to address the instability issue in value-based algorithms. Since the Actors of MADDPG output continuous value actions, Gumbel Softmax is applied to generate discrete actions, implement reparameterization, and control the exploratory tendency [28].

B. Motivation and Contributions

To address the aforementioned challenges and achieve efficient full frequency resource reuse in MB-LEO satellite systems, we propose a multi-agent actor-critic method for BH resource allocation, modeling the beam position selection and power allocation of each beam as an agent respectively, and allocating the beam time slot resources and power resources in two stages: Firstly, we focus on determining the beam pattern that maximizes channel capacity and system throughput. We guide the BH agents to balance maximizing throughput and minimizing packet loss by gradually adjusting the penalty of dumped traffic requests during the training phase, which helps in achieving BH pattern that most beams are positioned at a far distance from others, thus reducing inter-beam interference effectively under full-frequency reuse. Then we allocate the power of each beam by Power Allocation (PA) agent to ensure that the channel capacity optimized matches the traffic demand to maximize resource utilization. This approach allows for the dynamic allocation of time slots and power resources on demand and avoids interference between beams effectively.

Our main contributions are summarized as follows.

- We propose a comprehensive framework and its mathematical formulation to support efficient BH resource allocation in MB-LEO satellite systems through a multi-agent actor-critic method. In this framework, beam position selection and power allocation are modeled as agents, allowing for dynamic and flexible resource management.
- To tackle the multi-objective, multi-dimensional optimization problem, the resource allocation process is divided into two stages: initially, the optimal beam pattern

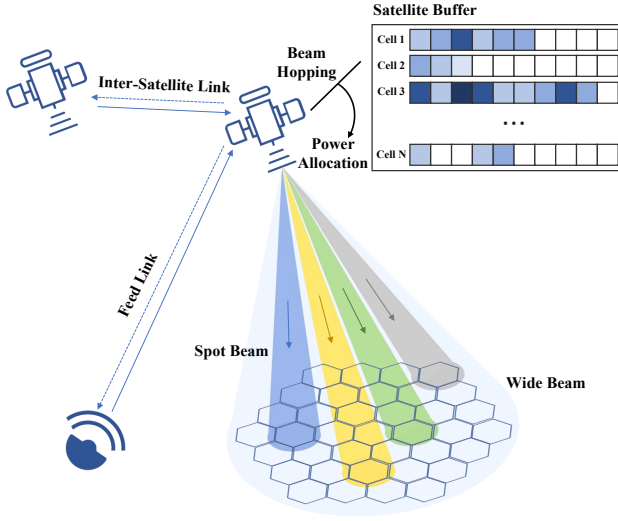


Fig. 1: Full frequency reuse MB-LEO satellite communication system.

is determined to maximize channel capacity and system throughput. Subsequently, a PA agent dynamically allocates power to each beam to match traffic demand and maximize resource utilization.

- We introduce a dynamic adjustment mechanism during the training phase that guides BH agents to balance throughput maximization and packet loss minimization by adjusting the penalty for dropped traffic requests. This mechanism ensures that beams are positioned to minimize interference effectively under full-frequency reuse conditions.
- Simulation results demonstrating that the proposed approach achieves 97% of the system's maximum transmission capacity, effectively reduces inter-beam interference, and maintains packet loss at less than 2%, indicating improved system throughput, reduced user queuing delay, ensured delay fairness, effective inter-beam interference management, and maximized resource utilization.

C. Organization

The remainder of this paper is structured as follows: Section II presents the system model and problem formulation, followed by the proposed two-stage optimization approach based on the multi-agent actor-critic method in Section III. Section IV. provides the simulation results and performance analysis. Finally, we conclude the paper in Section V.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. LEO Satellite Forward Link Model

In this paper, we focus on LEO satellites positioned at an orbital height of H_{orbit} . These satellites are equipped with phased array antennas, enabling them to dynamically steer multiple beams towards the ground and adjust their directions in real-time. The total power allocated to the satellite beams is

denoted as P_{total} , the total available bandwidth is represented by B_{total} , and the carrier frequency is denoted as f_c . We adapt a full frequency reuse scheme, where all beams share full bandwidth. Specifically, we concentrate on the forward link of the MB-LEO satellite communication system, as illustrated in Fig 1. The available on-board beam antennas are denoted as $K = \{k | k = 1, 2, \dots, K\}$, and the satellite provides services to N ground cells, denoted as $N = \{n | n = 1, 2, \dots, N\}$, where $K < N$. At each time slot, the MB-LEO satellite receives traffic requests from the ground cells and stores the untransmitted data in the satellite traffic buffer, denoted as $D = \{d_t^{n,l} | n \in N, l \in [1, T_{\text{ttl}}]\}$. Here, d_t^n represents the traffic of cell n stored in the buffer, which has been waiting for l time slots. T_{ttl} is the maximum waiting time for traffic, and any unserved traffic exceeding T_{ttl} will be discarded, resulting in packet loss. In time slot t , the satellite selects up to K cells to provide services and allocate power to each beam. The selected cell of each beam is denoted as $I = \{I_t^k | k \in K, I_t^k \in N\}$, where I_t^k represents the cell number illuminated by beam k . The overall beam pattern is denoted as $S = \{s_t^n | n \in N, s_t^n \in \{0, 1\}\}$, where s_t^n indicates whether cell n is illuminated by any beam. The power allocation of beams is denoted as $P = \{p_t^k | k \in K\}$.

During transmission on the forward link, our primary consideration is the free space loss, which can be calculated using the following formula:

$$\text{FSL} = 32.44 + 20 \log_{10}(f_c) + 20 \log_{10}(d), \quad (1)$$

where d is the distance between the satellite and the ground cell. The channel gain from the beam k to the cell n can be calculated by:

$$h_k^n = G_T + G_R - \text{FSL}, \quad (2)$$

where G_T and G_R represent the satellite transmitting antenna gain and ground receiving antenna gain.

As a full frequency reuse scheme is adopted for the MB-LEO satellite in this paper, resulting in mutual interference between adjacent beams within the illuminated areas. The signal-to-interference plus noise ratio (SINR) of cell n can be calculated using the following formula:

$$\text{SINR}_t^n = \frac{p_t^{(n,k)} h_k^n}{N_0 B_{\text{total}} + \sum_{|n' - n| \leq 4R} \sum_{k' \neq k}^K p_t^{(n',k')} h_{k'}^{n'} \psi^{|n' - n|}}, \quad (3)$$

where $p_t^{(n,k)}$ represents the power allocated to beam k , which serves the cell n in time slot t , and $p_t^{(n,k)} h_k^n$ is the power of the useful signal at the receiver. $N_0 B_{\text{total}}$ represents the environmental noise, and the interference caused by beams serving adjacent cells is calculated by right half of the denominator, where $|n' - n| \leq 4R$ indicates that the distance between the beam center points is less than or equal to 4 cell sides, that is, the mutual interference is calculated only when the two beams are located in adjacent cells or the same cell, $\psi^{|n' - n|}$ is beam power attenuation with spacing.

This paper considers that satellite spot beams project onto the ground as circles of equal radius. Under full frequency

reuse conditions, overlapping or nearly overlapping beam positions cause interference from adjacent beam side lobes, significantly affecting the signal-to-noise ratio and reducing channel capacity. The interference intensity between beams is inversely related to the spacing of their positions. For simplification, we assume the main lobe of the MB satellite's spot beam covers a ground cellular cell, with the main lobe width equal to the cell radius and the 3 dB beam coverage width equal to 2 cell radii. When the distance between two beam centers exceeds 3 cell radii, interference from side lobe overlap is significantly reduced. Therefore, the distance threshold for spatial isolation between beams is 3 cell radii.

B. Evaluation Criteria of LEO satellite systems

In satellite communications, the evaluation of BH service quality and performance typically includes throughput, packet loss, average delay, and delay fairness among all cells.

According to Shannon's Theorem, the channel capacity of cell n in time slot t can be calculated as follows:

$$C_t^n = \begin{cases} B_{\text{total}} \log_2(1 + \text{SINR}_t^n), & \text{cell } n \text{ is illuminated} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

The throughput of cell n in time slot t is determined by the minimum value between the channel capacity and the total awaiting traffic:

$$\text{Th}_t^n = \min\{C_t^n, D_t^n\}, \quad (5)$$

where D_t^n is the total awaiting traffic of cell n in time slot t , and can be calculated by:

$$D_t^n = \sum_{l=1}^{T_{\text{ttl}}} d_t^{n,l}. \quad (6)$$

The unserved traffic will be discarded if its waiting time exceeds T_{ttl} . The packet loss of cell n in time slot t can be calculated by:

$$L_t^n = \min\{C_t^n, d_{T_{\text{ttl}}}^n\}. \quad (7)$$

The average delay of traffic of cell n in time slot t can be calculated as follows:

$$\text{Delay}_t^n = \frac{\sum_{l=1}^{T_{\text{ttl}}} l \cdot d_t^{n,l}}{\sum_{l=1}^{T_{\text{ttl}}} d_t^{n,l}}. \quad (8)$$

Thus, we define the system average delay and delay fairness as:

$$\text{AverageDelay} = \frac{\sum_{t \in T, n \in N} \text{Delay}_t^n}{T \cdot N}. \quad (9)$$

The system delay fairness index is used to measure the fairness of beam services in high-demand areas and remote areas. High-demand areas can be continuously covered by satellite beams, so their delay is lower than that of occasional demand areas with less beam coverage. The calculation formula of the system delay fairness index is:

$$\text{DelayFairness} = \frac{\sum_{t \in T} (\max_{n \in N} \{\text{Delay}_t^n\} - \min_{n \in N} \{\text{Delay}_t^n\})}{T}. \quad (10)$$

After obtaining the BH pattern, the MB-LEO satellite allocates power to each beam. The power utilization of beam k in time slot t is calculated as the ratio between the transmitted traffic of the illuminated cell and the channel capacity provided by the beam. Therefore, the system power resource utilization efficiency is:

$$u_t^k = \sum_{n=1}^N \frac{\text{Th}_t^n}{C_t^n}. \quad (11)$$

Among the five indicators, total channel capacity serves as the foundational performance limit, as it directly reflects the mutual interference between beams. If same-frequency beams illuminate the same or adjacent cells, the SINR in the channel capacity calculation will drop significantly, reducing this capacity. System throughput, which is fundamentally constrained by the available channel capacity, reflects how well the cell's service requests are met. When throughput approaches total service demand, the algorithm has found an effective beam-hopping strategy. Packet loss, a direct consequence of throughput failing to meet demand over time, is a better indicator of the BH strategy's flexibility and fairness compared to system throughput. Continuous illumination of a cell by a beam may ignore surrounding cells' service requests to avoid interference, and remote, low-demand areas may be overlooked. Thus, optimizing packet loss requires a trade-off factor. Similarly, average user delay is impacted by service queues, while delay fairness is crucial for characterizing how equitably this performance is distributed to ensure a good user experience.

C. Problem Formulation

In satellite communications, the evaluation of BH service quality and performance typically includes throughput, packet loss, and delay. Among these metrics, packet loss has a more significant impact on user experience compared to limited delay. The optimization problem is formulated as follows:

$$\text{opt. } \mathcal{P}_1 = \max \sum_{t \in T} \sum_{n \in N} \text{Th}_t^n, \quad (12)$$

$$\mathcal{P}_2 = \min \sum_{t \in T} \sum_{n \in N} L_t^n, \quad (13)$$

$$\mathcal{P}_3 = \max \sum_{t \in T} \sum_{k \in K} u_t^k, \quad (14)$$

$$\text{s.t. } \mathcal{C}_1 = \sum_{k=1}^K p_t^k \leq P_{\text{total}}, \quad (15)$$

$$\mathcal{C}_2 = p_t^k \leq P_{\text{max}}, \quad (16)$$

$$\mathcal{C}_3 = \sum_{n=1}^N s_t^n \leq K, \quad (17)$$

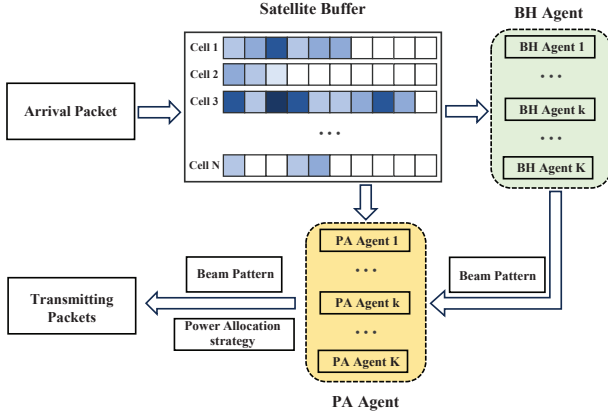


Fig. 2: BH resource allocation process based on Multi-Agent reinforcement learning.

where $\mathcal{P}_1$ aims to maximize system throughput under the condition of full frequency reuse. It encourages maximizing channel capacity by minimizing mutual interference between adjacent cells. $\mathcal{P}_2$ focuses on minimizing packet loss caused by uneven traffic distribution. This ensures that the delay in most high-demand cells will be below T_{ttl} and prevents low-demand cells from having their requests ignored. $\mathcal{P}_3$ ensures that the power allocated to each beam is determined based on actual needs, thereby maximizing power resource utilization. $\mathcal{C}_1$ indicates that the sum of the power allocated to all beams must not exceed the total available power. Constraint $\mathcal{C}_2$ sets an upper limit on the power allocated to each beam, ensuring that the power allocation remains within the bounds of $P_{\max}$. $\mathcal{C}_3$ indicates that there can be at most K ground cells illuminated by beams in each time slot.

III. TWO STAGE OPTIMIZATION BASED ON MULTI-AGENT ACTOR-CRITIC METHOD

In this section, we introduce a satellite resource allocation framework using a multi-agent reinforcement learning approach. Given the impracticality of simultaneously achieving the optimal solution for all three objectives, we adopt a two-stage approach. The first stage focuses on optimizing objectives $\mathcal{P}_1$ and $\mathcal{P}_2$ with the goal of maximizing channel capacity and throughput to meet the requirements of satellite communication while improving user QoS. In the second stage, we address the beam power allocation problem $\mathcal{P}_3$ to maximize the utilization of satellite power resources.

A. Overall process of the proposed solution

The overall process of the BH resource allocation algorithm based on multi-agent reinforcement learning (MARL) is shown in Fig. 2. After the MB satellite uses a wide beam to receive the ground cellular cell service requests, it forwards them through the constellation-ground gateway and stores the data to be transmitted in the onboard traffic buffer.

The beam position selection and power allocation decision of each beam of the MB satellite are modeled as a reinforcement learning agent. Therefore, the agents involved in

resource allocation are divided into two categories: BH Agent that determines the direction of the beam and PA Agent that determines the power value of the beam. The number of agents of each type is equal to the number of beams. The on-board data buffer is distinguished by the cell number and the waiting time length, and is vectorized and reconstructed as part of the agent network input for reference by the two types of agents when making decisions. Among them, the decision output of the BH Agent is a discrete value, representing the number of cells illuminated by the center of the beam. Then, the PA Agent allocates the corresponding power to each beam based on the beam diagram obtained by all beam hopping agents in the first stage to match the actual traffic load of the beam, and its decision output is a continuous value.

B. Beam Hopping Stage

In the first stage, power constraints are disregarded, and an equal power distribution is assumed among all beams. These beams adopt full frequency reuse, resulting in mutual interference between adjacent cells. Our objective is to collaboratively achieve a BH pattern that reduces inter-beam interference and maximizes satellite communication throughput using multi-agent actor critic method. This approach enables the transmission of a larger amount of communication data, ultimately fulfilling the requirements of each cell. Fig. 3 depicts the framework for training and execution. We employ centralized training with distributed execution paradigm, each beam is modeled as an agent, which consists of a decentralized actor network, a centralized critic network, and target networks of actor and critic.

1) *Observation of BH actor*: The observation of the actor network is derived from the satellite buffer, which serves as storage for the untransmitted traffic of each cell. It is represented as a matrix of size $[T_{ttl}, N]$. All actors share the same observation: a vector of size $[T_{ttl} \times N, 1]$ obtained by expanding the satellite buffer, where each element of the list represents the normalized size of the waiting traffic of cell n with the waiting time of l .

$$o_t = \left[\frac{d_t^{n,l}}{d_{\max}^{n,l}} \right], \quad n \in N, l \in T_{ttl}, \quad (18)$$

$$o_t^1 = o_t^2 = \dots = o_t^K = o_t, \quad (19)$$

where $d_{\max}$ is a constant representing the maximum value of the traffic to normalize the observations.

2) *Action of BH actor*: The actor network outputs a probability distribution of action values with a length of N , which is suitable for generating continuous actions. However, directly selecting the maximum value using the $\arg\max$ function as the discrete action is not feasible due to its non-differentiability. To generate discrete actions and allow the network to train, the Gumbel-Softmax approach is applied, which provides a continuous and differentiable approximation of the discrete categorical distribution, enabling the use of gradient-based optimization algorithms for training models involving discrete choices or categorical variables, overcoming the challenge posed by non-differentiable $\arg\max$ functions. The discrete action of actor network is a one-hot vector:

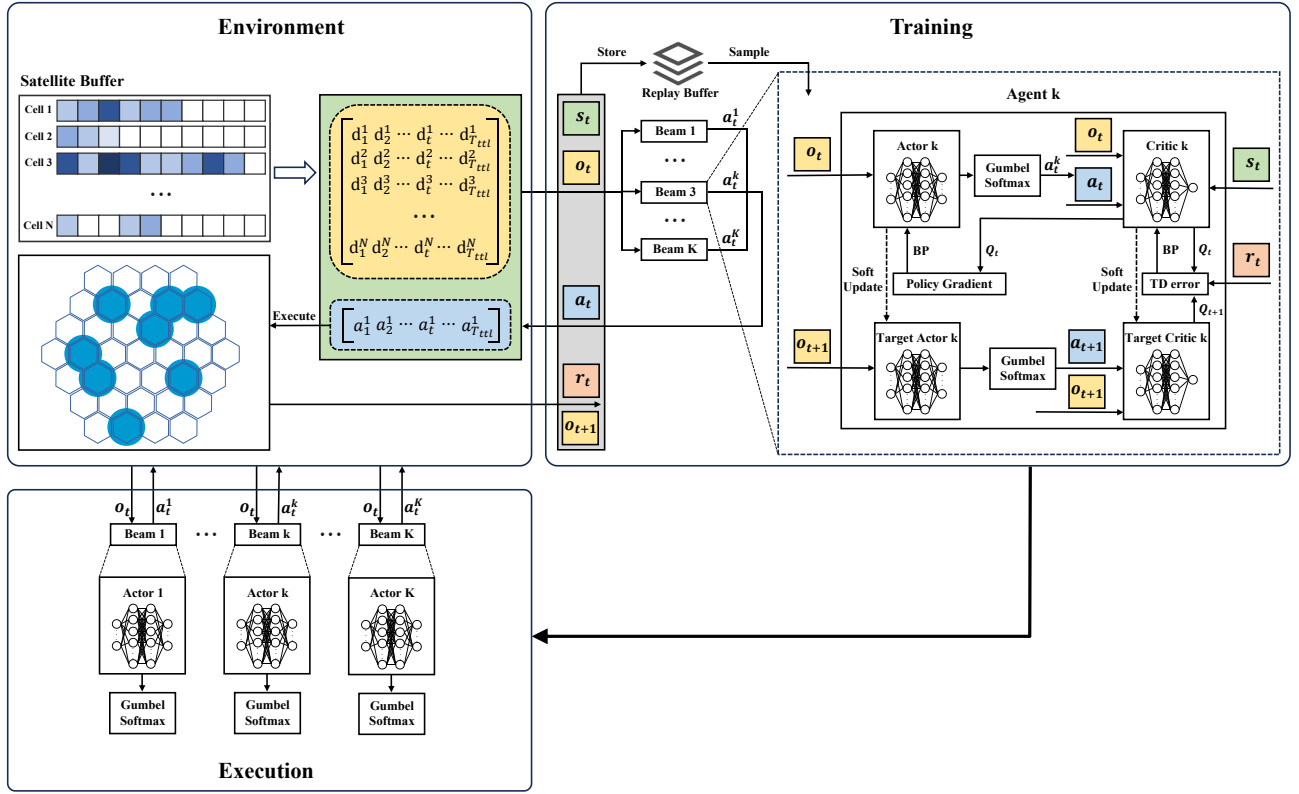


Fig. 3: Proposed MAAC-GS algorithm for training and execution in BH stage.

$$a_t^k = \text{onehot}[n], n \in N, \quad (20)$$

$$a_t = [a_t^1, a_t^2, \dots, a_t^K]. \quad (21)$$

3) *Observation of BH critic*: The advantage of the actor-critic method lies in its ability to introduce global observations during the training phase, allowing the agent to gain insights into the decision-making patterns of other agents. During the execution process, only a partial observation is required, specifically the satellite buffer o_t , to effectively avoid interference. This can be challenging to achieve in methods based on DQN. The observation of the critic network is the concatenation of the traffic information and one-hot vectors representing the actions of all agents:

$$s_t = [o_t, a_t], \quad (22)$$

$$s_t^1 = s_t^2 = \dots = s_t^K = s_t. \quad (23)$$

4) *Reward*: In the first stage, the main focus is to ensure that the communication needs of users are met, with the goal of maximizing traffic and minimizing packet loss. To achieve this, collaboration among multi-agents is necessary to enhance overall performance. Therefore, all agents receive the same reward:

$$r_t = \frac{\sum_{n=1}^N \text{Th}_t^n}{\text{Th}_{\max}} - \frac{\sum_{n=1}^N L_t^n}{\text{Th}_{\max} \cdot \beta}, \quad (24)$$

$$r_t^1 = r_t^2 = \dots = r_t^K = r_t, \quad (25)$$

where $\sum_{n=1}^N \text{Th}_t^n$ represents the total transmitted traffic at time t , $\sum_{n=1}^N L_t^n$ represents the total packet loss at time t , and $\text{Th}_{\max}$ is used for normalization. Additionally, β is introduced to control the penalty of packet loss. In the early stage of training, β is a constant, allowing the agent to primarily focus on learning cooperation strategies to maximize the overall system traffic. As the network gradually converges, β also decreases, increasing the penalty for untransmitted traffic, which encourages agents to prioritize the needs of reducing packet loss in low-traffic cells, thereby gaining better delay fairness for users.

5) *Parameter update*: As shown in Algorithm 1, the agents interact with the environment, store the obtained experience tuple $(o_t, a_t, s_t, r_t, o_{t+1})$ obtained at each time slot to the replay buffer, and sample a mini-batch from it during training. During the training phase, the critic network is updated through the Temporal Difference (TD) method, the loss function can be calculated as follows:

$$L(Q_{\Theta_k}) = \left[Q_{\Theta_k}(o_t, a_t) - (r_t + \gamma Q_{\Theta'_k}(o_{t+1}, \pi_{\Psi'}(o_{t+1}))) \right]^2, \quad (26)$$

where Q_{Θ_k} represents the value function of agent k , Θ_k and Θ'_k are the parameters of the critic and target critic network, respectively. Similarly, π_{Ψ_k} represents the policy function of agent k , Ψ_k and Ψ'_k are the parameters of the actor and target actor network, respectively. γ represents the discount factor. The joint action based on the observation o_{t+1} is obtained by

Algorithm 1: Training of BH agents with MAAC-GS algorithm.

Initialize:

- 1: Initialize actor and critic networks of K agents.
- 2: Hard-update the target actor and critic networks.
- 3: Initialize replay buffer.
- 4: Initialize temperature of Gumbel-Softmax.

Training:

- 5: **for** episode = 1 to $Warmup_Episode$ **do**
 - 6: Store the experience tuple (o_t, a_t, r_t, o_{t+1}) to replay buffer by executing random actions
 $a_t = (a_t^1, a_t^2, \dots, a_t^K)$.
 - 7: **end for**
 - 8: **for** episode = 1 to $Training_Episode$ **do**
 - 9: Reset the environment.
 - 10: **for** time_slot = 1 to $Episode_Length$ **do**
 - 11: Get observation o_t .
 - 12: For each agent, generate action a_t^k from the Actor network and Gumbel-Softmax, get
 $a_t = (a_t^1, a_t^2, \dots, a_t^K)$.
 - 13: Execute a_t , receive reward r_t and the next observation o_{t+1} .
 - 14: Store the experience tuple (o_t, a_t, r_t, o_{t+1}) to replay buffer.
 - 15: **end for**
 - 16: **for** train_time = 1 to $Train_Repeat$ **do**
 - 17: Sample mini-batch (o_t, a_t, r_t, o_{t+1}) .
 - 18: Calculate $Q_{\Theta}(o_t, a_t)$ by Critic.
 - 19: Calculate a_{t+1} from $\pi_{\Psi'}(o_{t+1})$ by Target Actor.
 - 20: Calculate $Q_{\Theta'}(o_{t+1}, a_{t+1})$ by Target Critic.
 - 21: Update Critic according to (26).
 - 22: Calculate a^k from $\pi_{\Psi}(o_t)$ by Actor.
 - 23: Calculate $Q_{\Theta}(o_t, a_t^1, \dots, a_t^k, \dots, a_t^K)$ by Critic.
 - 24: Update Actor according to (27).
 - 25: Update target network by soft update method.
 - 26: **end for**
 - 27: **if** performance no longer improves **do**
 - 29: Clear replay buffer, decay β , warmup based on the trained model.
 - 30: **end if**
 - 31: **end for**
-

$\pi_{\Psi'}(o_{t+1})$.

The actor network updates through the Policy Gradient method, and the policy gradient is as follows:

$$\nabla_{\Psi_k} J(\pi_{\Psi_k}) = \mathbb{E}_{\pi_{\Psi_k}} [\nabla_{\Psi_k} \log(\pi_{\Psi_k}(a_t^k | o_t)) Q_{\Theta_k}(o_t, a_t)], \quad (27)$$

where $\pi_{\Psi_k}(a_t | o_t)$ represents the probability of a_t^k under the observation of o_t . After each training, the target networks of the actor and critic are updated through soft update method.

In the training phase, the agents utilize critic networks to estimate the values of the current state, and the policy functions of the actors are adjusted to output actions that maximize the estimated values. As a result, the actor network demonstrates lower sensitivity to the reward value than that of

the DQN method, which enables us to exert control over the learning process of the agents by adjusting the penalty factor β , to achieve a BH pattern that balances maximizing system throughput and minimizing packet loss gradually.

6) *Distributed execution:* After centralized training, the actor network of the agent forms its own decision-making mode and can effectively avoid overlapping or adjacent service areas. In the distributed execution mode, the trained agent only needs to observe the status of the satellite traffic buffer, and the actor network makes independent decisions and outputs the decision results of the beam position through the GS estimator. The following will introduce in detail the techniques of using the GS estimator to generate discrete actions.

7) *Generation of discrete actions-GS reparameterization:* The actor network uses the policy gradient algorithm to update the network parameters. It determines the beam position decision result in the current environment by observing the traffic demand state o_t of the local environment. Its network output layer generates a one-dimensional action probability distribution with a length of N :

$$\text{logits}_t = [\text{logits}_t^n \mid n \in N]. \quad (28)$$

In the problem of obtaining discrete outputs of the Actor network, if the Softmax operation is used and the number of the maximum value is selected as the decision result, the action probability distribution of the network output layer will lose its probability meaning due to Softmax scaling, and can only reflect that a certain action has a higher tendency to be selected; if a discrete action value is obtained by directly sampling the probability distribution based on logits, the meaning of the action selection probability distribution is retained, but this action value causes the gradient to be lost, and the agent network cannot be trained through the back propagation (BP) algorithm. Reference [28] discusses the problem of applying the GS estimator in the multi-agent deep deterministic policy gradient algorithm. The GS estimator can not only reflect the meaning of the probability distribution of the policy function in action selection, but also retain the gradient to train the network. Therefore, the GS estimator is used in this algorithm and reparameterization is performed.

First, we generate independent samples that obey uniform distribution $[\mathcal{Y}_1, \mathcal{Y}_2, \dots, \mathcal{Y}_N]$, and then calculate Gumbel noise by taking negative logarithm:

$$G_n = -\log(-\log(\mathcal{Y}_n)), \quad (29)$$

We then add the noise G_i to the actor network output logits_t to get the new vector $[\text{logits}_t + G_n \mid n \in N]$. We calculate the probability distribution of each action through the Softmax function with temperature coefficient control as follows:

$$w_{\tau}(v_n) = \frac{e^{v_n/\tau}}{\sum_{n=1}^N v_i/\tau}, \quad (30)$$

where τ is the temperature coefficient, which is used to control the smoothness of the output distribution, that is, the amplification effect of the Softmax function on high-probability actions.

When τ is large, this distribution is less different from the original probability distribution, which can retain the meaning of the action probability distribution function and can also prompt the agent to explore a larger range of actions, which is suitable for the initial stage of agent training; as the agent gradually learns, the value of τ is gradually reduced, so that better actions with higher probabilities are more likely to be selected. The probability distribution calculated by Softmax is recorded as a soft decision as follows:

$$y_{\text{soft}} = [\omega_{\tau}(v_n) \mid n \in N]. \quad (31)$$

The action probability distribution y_{soft} still retains the gradient, selects the number of the largest item in y_{soft} , and forms a vector in one-hot encoding form as the beam position hard decision result:

$$y_{\text{hard}} = \text{onehot}[\arg\max_N(y_{\text{soft}})]. \quad (32)$$

The algorithm uses discrete actions of hard decisions in forward propagation and gradients of soft decisions in back propagation, that is, the gradient of y_{hard} will be replaced by the gradient of y_{soft} , so that the entire decision optimization problem becomes differentiable and can be optimized by the gradient descent algorithm.

C. Power Allocation Stage

In the second stage, power resources allocation is based on the beam diagram of BH time slot allocation to determine the power value allocated to each beam. Therefore, the BH agent is also part of the power resource allocation environment with a MADDPG structure.

Despite efforts to avoid inter-beam interference, the uneven distribution of traffic demand may still lead to conflicts between adjacent beams. With full frequency reuse, each beam's channel capacity is not solely determined by its own power but also influenced by the power of beams illuminating the surrounding cells. Our objective is to maximize resource utilization in the satellite communication system and efficiently allocate power to each beam based on demand while considering power constraints and interference between beams.

Different from the discrete value operation of time slot resources, the allocation result of beam power is a continuous value. Therefore, the MADDPG algorithm suitable for continuous actions is adopted. Similar to the MAAC-GS algorithm structure, in the beam power resource allocation scenario, the power decision of each beam is modeled as an agent, and the Actor-Critic structure and target network technology are also adopted.

1) *Observation of PA actor*: The observations of the actor networks in PA agents include both the satellite traffic buffer and the one-hot actions of BH agents as follows:

$$o_t^p = s_t = [o_t, a_t]. \quad (33)$$

2) *Action of PA actor*: Each actor network outputs a continuous value ranging from -1 to 1. We convert the output to $[0, P_{\max}]$, thus each dimension in the joint action vector corresponds to the power allocated to the beam k .

$$p_t = [p_t^1, p_t^2, \dots, p_t^K], \quad p_t^k \in [0, P_{\max}]. \quad (34)$$

The value critic network outputs the expected return PAQ_t of the power allocation result under a given beam hopping mode. The network has global observations and its input is:

$$pa_{st} = [o_t, a_t, p_t]. \quad (35)$$

3) *Reward*: The power resource allocation stage mainly considers the optimization goal $\mathcal{P}_3$, that is, according to the service demand of the cell illuminated by the beam, the power resources that match the beam are allocated to make the channel capacity provided by the beam no higher than the service demand, thereby improving resource utilization efficiency. The reward function for PA agent k is defined to maximize overall power utilization while adhering to the limitation of total available power:

$$r_t^k = \begin{cases} -\frac{|D_t^{a_k} - C_t^{a_k}|}{D_{\max}}, & \sum_{k=1}^K p_t^k \leq P_{\text{total}}, \\ -10, & \text{otherwise} \end{cases}, \quad (36)$$

here, each agent is rewarded separately, and all agents receive a penalty if the total allocated power exceeds the total power limit.

4) *Parameter update*: The update method of the MADDPG algorithm model network is consistent with the update method of the MAAC-GS algorithm. The value network is updated using the temporal difference learning algorithm, and the policy network is updated using the policy gradient algorithm as shown in Algorithm 2. At the beginning of training, the PA agent does not know the number of the beam corresponding to itself. It learns the corresponding beam number by continuously learning from the feedback rewards of the environment. Then, in the search for strategies that can reduce penalties, it first learns how to cooperate with other agents to avoid the large penalty items of environmental feedback due to the total power exceeding the limit. Then gradually improve the ability of the beam channel capacity to match business needs to reduce resource waste and achieve optimization of the target $\mathcal{P}_3$.

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

In this section, we perform simulations to evaluate the effectiveness of our proposed framework. The simulation settings and parameters are presented in Table I. Subsequently, we provide a comprehensive analysis and explanation of the obtained simulation results.

A. Simulation Settings

We consider the MB-LEO satellite positioned at an orbital height of 550 km, employing 8 spot beams to provide services to 37 cells. Our system assumes full frequency reuse, where

Algorithm 2: Training of Power allocation agent

Initialize:

- 1: Initialize actor and critic network of PA agent.
- 2: Hard-update the target actor and critic networks.
- 3: Initialize replay buffer.

Training:

- 5: **for** episode = 1 to *Warmup_Episode* **do**
- 8: Store transition $(\{o_t, a_t\}, p_t, r_t, \{o_{t+1}, a_{t+1}\})$
to replay buffer by executing random action
 $p_t = (p_t^1, p_t^2, \dots, p_t^K)$.
- 9: **end for**
- 10: **for** episode = 1 to *Training_Episode* **do**
- 11: Reset the environment.
- 12: **for** time_slot = 1 to *Episode_Length* **do**
- 13: Get observation $\{o_t, a_t\}$.
- 14: For each agent, generate action a_t^k from the
Actor network, get $p_t = (p_t^1, p_t^2, \dots, p_t^K)$.
- 15: Execute p_t , receive reward r_t and the next
observation $\{o_{t+1}, a_{t+1}\}$.
- 16: Store the experience tuple
 $(\{o_t, a_t\}, p_t, r_t, \{o_{t+1}, a_{t+1}\})$.
- 17: **end for**
- 18: **for** train_time = 1 to *Train_Repeat* **do**
- 19: Sample mini-batch
 $(\{o_t, a_t\}, p_t, r_t, \{o_{t+1}, a_{t+1}\})$.
- 20: Calculate $Q_\Theta(\{o_t, a_t\}, p_t)$.
- 21: Calculate p_{t+1} from $\pi_{\Psi'}(o_{t+1}, a_{t+1})$.
- 22: Calculate $Q_{\Theta'}(\{o_{t+1}, a_{t+1}\}, p_t)$.
- 23: Update Critic according to (26).
- 24: Calculate p'_t from $\pi_\Psi(o_t, a_t)$.
- 25: Calculate $Q_\Theta(\{o_{t+1}, a_{t+1}\}, p'_t)$.
- 26: Update Actor according to (27).
- 27: Update target network by soft update
method.
- 28: **end for**
- 29: **end for**

all beams share the same bandwidth. The transmission quality at the receiving end is influenced by environmental noise and mutual interference between adjacent beams with the same frequency. If demand waiting to be transmitted exceeds a delay threshold of 10 time slots, the transmission is unsuccessful, resulting in packet loss.

For the BH performance comparison, 7 schemes are considered in the simulation part:

- *The Random Beam Hopping (Random)* : Randomly determine the cell to serve for each beam [29].
- *The Greedy Beam Hopping (Greedy)* : Perform K selections consecutively by choosing the cell with the highest traffic demand at each time, and set the demand of the chosen cell to zero [30].
- *The Periodic Beam Hopping (Periodic)* : Select the BH pattern by the periodic service queue. In this scheme, the 8 beams sequentially serve the 37 cells once every 5 time slots [26].
- *The Genetic Algorithm Beam Hopping (Genetic)* :

Generate the BH pattern by the genetic algorithm, where the initial population is 200 and the number of genetic iterations is 20 [31].

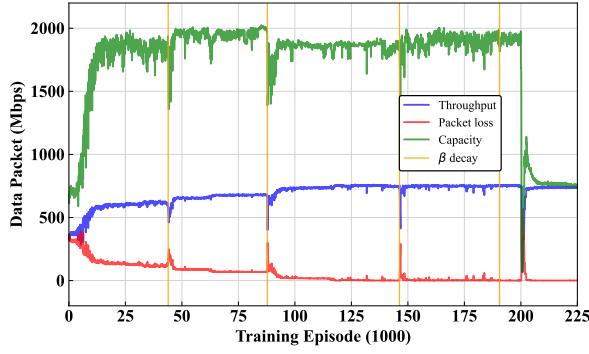
- *The DQN Beam Hopping (Single)* : According to [21], the DQN method is applied, where the single agent selects K cells respectively through sequential decision in each time slot.
- *The CommNet Beam Hopping (CommNet)* : CommNet is a communication-based multi-agent algorithm designed to facilitate multi-agent cooperation by enabling multiple communications between RNN hidden layers. It has proven effective in environments such as MaZeBASE, which demands collision avoidance [32]. In the context of the BH problem, the CommNet agents observe the traffic buffer and generate an action of size K to determine the illuminating cells of beams.
- *The QMIX Beam Hopping (QMIX)* : QMIX is a value decomposition based multi-agent algorithm. It assigns different weights to each agent's contribution to the collective result by a super network which estimates the Q value of joint actions as a complex nonlinear combination of each agent's Q value [33]. The observation and global state of agents in QMIX are the same with the approach we proposed. The QMIX model we used for evaluating was trained for 20000 episodes.

TABLE I: Simulation Parameters

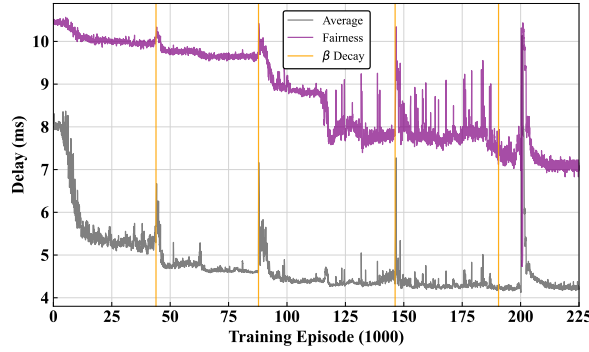
PARAMETERS	VALUES
ENVIRONMENT PARAMETERS	
Ku-band f_c	11.7GHz
Bandwidth available B	250MHz
Beam power available P	13dBW
Beam power limit P_{max}	10dBW
Number of cells N	37
Number of beams K	8
Orbit altitude H_{orbit}	550km
Transmitting antenna gain G_T	40.3dBi
Receiving antenna gain G_R	31.6dBi
Noise power spectral density N_0	-151.1dBm/Hz
Total Traffic Demand	750Mbps
Cell Traffic	$U(0Kb, 81Kb)/timeslot$
Power attenuation in adjacent cell ψ	0.5
Beam hopping dwell time $T_{timeslot}$	2ms
Maximum waiting time T_{ttl}	$10 \times T_{timeslot}$
Episode time slots	50
BEAM HOPPING PARAMETERS	
Training episodes	200000
Hidden layer	2
Hidden layer dimension	128
Batch size	128
Learning rate	0.005
Discount factor γ	0.99
Soft update rate	0.01
Penalty factor β	1.0
POWER ALLOCATION PARAMETERS	
Training episodes	25000
Hidden dimension	512
Batch size	512
Learning rate	0.001

B. System indicators during training

The training process of BH agents and PA agents is illustrated in Fig. 4. In the initial stage of training, the BH agents gradually learn to cooperate to serve the cells while keeping



(a) System throughput, packet loss and channel capacity.



(b) Average delay and delay fairness.

Fig. 4: Performance of BH agents during training.

a far distance, thereby providing more channel capacity. The rapid growth of channel capacity ahead of throughput indicates that interference avoidance is the foundational policy learned by the agents. But the agents were less capable of avoiding packet loss. We increase the penalty of packet loss by decaying β , which discourages the agents from prioritizing areas with high traffic only. It can be observed that after each decay of β , there is a reduction in packet loss, along with significant improvements in average delay and delay fairness. The well-trained BH agents achieve an average delay of 4.2 ms while meeting most of traffic demand, and reduce the delay fairness to 7.0 ms. The training of PA agents started from episode 20000. During the PA training phase, the channel capacity provided by beams gradually approaches the amount of traffic that needs to be transmitted. Eventually, the channel capacity becomes nearly equal to the throughput without compromising system performance like packet loss and delay. This implies that the power of each beam was efficiently allocated according to the demand. These results highlight the effectiveness of our approach in reducing interference, ensuring efficient resource utilization and enhancing overall communication performance.

C. Beam patterns generated by the BH agent

Fig. 5 shows the BH patterns generated by BH agents in three consecutive time slots. The depth of red represents the total traffic demand of each cell, while the blue color

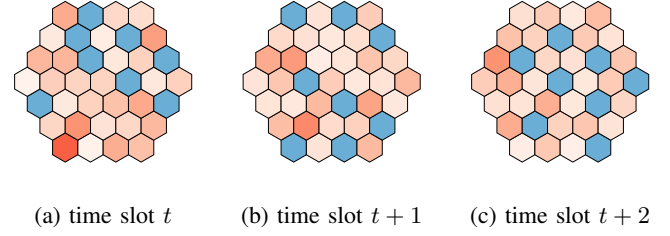


Fig. 5: BH pattern generated by BH agents.

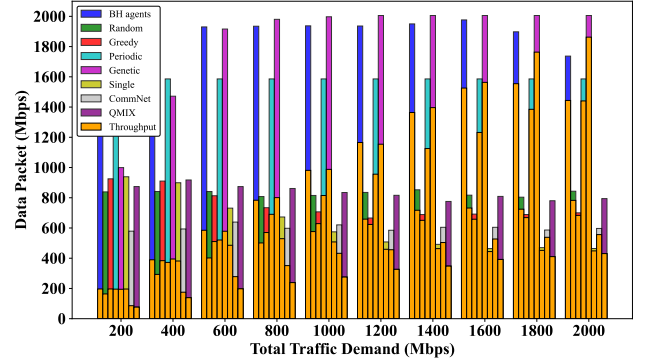


Fig. 6: System channel capacity of different BH schemes under varying traffic demand.

represents the cells illuminated by beams. It is evident that cells with high demand are served promptly. This BH method avoids the occurrence that one area is served by multiple beams, and there is a certain distance between the illuminated areas of each beam, which effectively avoids interference. After training on ground servers, the learned parameters of actors are transmitted to the satellites via secure and reliable communication links [34] [35].

D. Comparison and analysis

To evaluate the performance of the MAAC-GS-based BH time slot resource allocation algorithm proposed in terms of system throughput and interference avoidance, the condition that the beam power is evenly distributed to each beam is used, and a total of 7 BH schemes are used for performance comparison.

1) *System channel capacity under varying traffic demand:* In Fig. 6, we evaluate the performance of different BH schemes under varying total traffic demand, ranging from 200Mbps to 2000Mbps. To compare the BH performance, we assumed that the total beam power was evenly distributed to each beam. The results show that our proposed method and the genetic algorithm BH perform significantly better than other schemes. The proposed method achieved high channel capacity and effectively transmits most of the traffic across different total demand scenarios, highlighting its efficiency in avoiding interference and providing services. When the traffic demand is in the range of 600Mbps to 1600Mbps, the throughput and channel capacity of the proposed method are slightly lower than those of the genetic algorithm, and

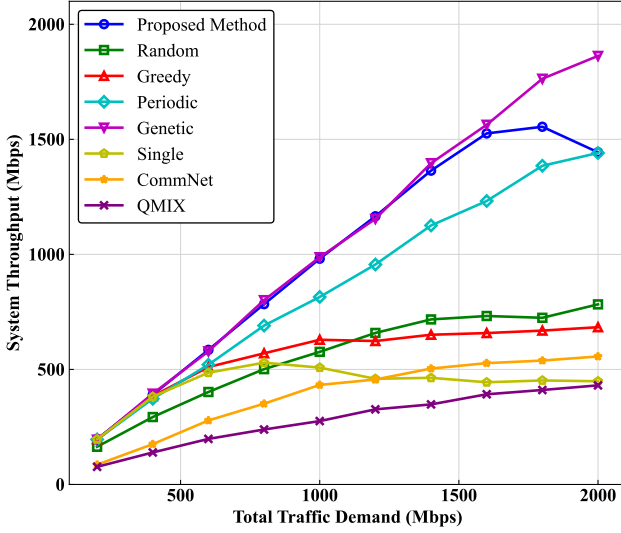


Fig. 7: System throughput of different BH schemes under varying traffic demand.

the performance of our algorithm degrades when the traffic demand is close to the maximum channel capacity. However, our approach is more resource-efficient, avoiding unnecessary power waste when the traffic demand is under 600 Mbps. Furthermore, in terms of decision-making efficiency, the average total inference time of all BH actor networks is about 3.2ms, while the genetic algorithm takes about 1611.8ms to generate an action. Such a substantial tendency is unacceptable for real-time BH decision-making.

2) *System throughput under varying traffic demand:* Fig. 7 shows the system throughput performance of the proposed algorithm compared to seven other algorithms under various total cell service loads, ranging from 200 Mbps to 2 Gbps. The results indicate that as total service demand increases, the throughput of all algorithms, except the DQN algorithm, improves. Notably, the time slot resource allocation algorithm based on MAAC-GS and the genetic algorithm outperform other beam-hopping modes.

In the range of [200, 1600] Mbps, the performance curves of these two algorithms resemble a positive proportional function with a slope of 1, indicating effective transmission. However, when total traffic demand exceeds 1600 Mbps, the performance of the MAAC-GS-based algorithm declines. This is because the model was primarily trained with an average total traffic load of approximately 750 Mbps. Consequently, its learned policy for maximizing throughput is highly optimized for typical loads but was not sufficiently explored for the extreme high-load regime. Addressing this limitation by training the model on a more diverse range of traffic scenarios is a key area for future work. Even with this decline, our proposed algorithm remains superior to other algorithms except for the genetic algorithm and polling scheduling. The genetic algorithm excels by finding the optimal solution in the search space of genetic iteration, adapting to various service demands. Polling scheduling benefits from pre-defined beam patterns that avoid inter-beam interference through spatial isolation,

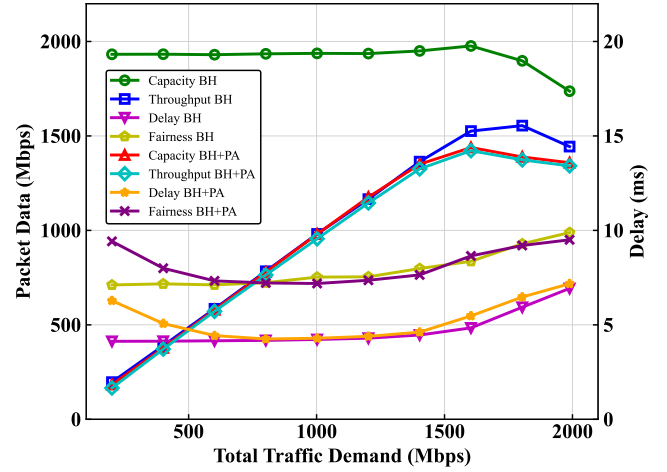


Fig. 8: System throughput and channel capacity of BH and BH with PA.

achieving high throughput through periodic access.

However, under low total demand, polling scheduling underperforms compared to the MAAC-GS-based algorithm. This is because polling scheduling accesses each cell only once every 5 time slots, potentially causing packet loss in high-demand cells before the next access. The proposed algorithm can continuously serve high-demand areas, demonstrating the excellent performance and flexibility of the MAAC-GS-based time slot resource allocation algorithm under typical service loads.

3) *System throughput and channel capacity of BH and BH with PA:* Fig. 8 compares the communication performance of the BH only scheme and the BH with PA scheme under varying traffic demands. For total traffic demands between 200 Mbps and 1400 Mbps, the channel capacity in the BH with PA scheme nearly matches the throughput of both schemes, indicating efficient beam power distribution by the PA agents based on required channel capacity.

When the demand ranges from 200 Mbps to 400 Mbps, the average delay of the BH with PA scheme is 1.5 ms higher than the BH scheme. Additionally, with demands from 1400 Mbps to 2000 Mbps, the BH with PA scheme's throughput is on average 9.3% lower than the BH scheme due to the channel capacity limitations.

E. Algorithm complexity analysis

In order to evaluate the practical applicability of our proposed method, we analyzed the computational complexity of the proposed two-stage resource allocation framework, including both the training and inference phases. Table II summarizes the computational complexity and execution time of the proposed method compared to benchmark approaches.

1) *Complexity of the Proposed Algorithm:* The MAAC-GS-based BH algorithm involves K agents, during the training phase, each agent updates its actor and critic networks based on sampled experiences from the replay buffer according to 1. The critic network requires global observations, which results

TABLE II: Computational Complexity and Execution Time of Different Resource Allocation Schemes

Algorithm	Complexity	Decision Time
MAAC-GS BH	$O(K \cdot d_n \cdot d_{w_1}^2)$	3.2 ms
MADDPG PA	$O(K \cdot d_n \cdot d_{w_2}^2)$	2.7 ms
Two-stage	$O(K \cdot d_n \cdot (d_{w_1}^2 + d_{w_2}^2))$	6.2 ms
DQN	$O(K \cdot d_{n_{dqn}} \cdot d_{w_{dqn}}^2)$	5.2 ms
CommNet	$O(K^2 \cdot d_{w_{comm}}^2)$	16.7 ms
QMIX	$O(K \cdot d_{n_{qmixon}} \cdot d_{w_{qmixon}}^2)$	2.5 ms
Genetic	$O(P_{size} \cdot G_{iter} \cdot E_{env})$	1611.8 ms
Random	$O(K)$	57 us
Greedy	$O(K \cdot N \cdot T_{ttl} + K \cdot N \cdot \log N)$	89 us
Periodic	$O(1)$	21 us

in a complexity of $O((n_{obs} + n_{acts}) \cdot d_{w_1} + d_h \cdot d_{w_1}^2 + d_{w_1})$. The complexity of the actor network is $O(n_{obs} \cdot d_{w_1} + d_h \cdot d_{w_1}^2 + N \cdot d_{w_1})$, where n_{obs} is the size of the observation of the actor network that is equal to $N \cdot T_{ttl}$, n_{acts} is the size of a one-hot vector containing the actions of all agents that is equal to $K \cdot N$, d_{w_1} is the dimension of hidden layers, and d_h is the number of hidden layers. And each agent performs a soft update for its target networks. This involves a weighted average of the parameters, with complexity $O(d_{w_1}^2)$ per agent. Considering K agents, the overall complexity for one training step is:

$$O(K \cdot ((2NT_{ttl} + NK + N + 1) \cdot d_{w_1} + 2d_n \cdot d_{w_1}^2)), \quad (37)$$

during the inference phase, only the Actor networks are used to generate actions for each agent. The complexity for all actor networks is:

$$O(K \cdot ((NT_{ttl} + N) \cdot d_{w_1} + d_n \cdot d_{w_1}^2)). \quad (38)$$

The MADDPG-based PA algorithm also involves K agents. Similar to the BH stage, the the training complexity is:

$$O(K \cdot ((2NT_{ttl} + 2NK + K + 2) \cdot d_{w_2} + 2d_n \cdot d_{w_2}^2)), \quad (39)$$

and the overall inference complexity is:

$$O(K \cdot ((NT_{ttl} + NK + 1) \cdot d_{w_2} + d_n \cdot d_{w_2}^2)). \quad (40)$$

The training complexity is higher due to the centralized training paradigm and parameter updating, but is performed offline and does not impact real-time operations. Since the dimension of the hidden layers dominates in the above formulas, the combined inference complexity for the BH and PA stages can be simplified as:

$$O(K \cdot d_n \cdot (d_{w_1}^2 + d_{w_2}^2)). \quad (41)$$

The inference phase of the proposed approach relies solely on the actor networks, which consist of simple multi-layer fully connected structures with a relatively small number of parameters and minimal computational requirements, primarily involving basic matrix multiplications. This lightweight design ensures that the model operates efficiently within the processing capabilities of onboard satellite processors, making it well-suited for deployment in resource-constrained satellite environments.

2) *Comparison with Benchmark Algorithms:* Neural network-based methods DQN and the Q-network of QMIX shared similar structure with the actors of MAAC-GS and MADDPG, therefore the computational complexity mainly depends on the dimensions of the hidden layers, and were similar to our proposed approach. Specifically, the complexity of DQN is $O(K \cdot d_{n_{dqn}} \cdot d_{w_{dqn}}^2)$, where $d_{w_{dqn}}$ is the dimension of hidden layers of DQN, and $d_{h_{dqn}}$ is the number of hidden layers. The complexity of QMIX is $O(K \cdot d_{n_{qmixon}} \cdot d_{w_{qmixon}}^2)$, where $d_{w_{qmixon}}$ is the dimension of hidden layers of Q-network, and $d_{h_{qmixon}}$ is the number of hidden layers. CommNet introduces communication between agents, resulting higher computational complexity of $O(K^2 \cdot d_{w_{comm}}^2)$, where $d_{w_{comm}}$ is the dimension of its hidden layers.

Traditional methods, such as random action selection, periodic action execution, and greedy BH, have lower computational complexity compared to neural network-based approaches. However, these methods lack adaptability, which leads to suboptimal system throughput and ineffective interference management. The genetic algorithm employs an iterative evolution process over a population of solutions, resulting in a significantly higher computational complexity of $O(P_{size} \cdot G_{iter} \cdot E_{env})$. Here, P_{size} represents the population size, G_{iter} is the number of generations, and E_{env} denotes the evaluation complexity of the traffic updating procedure, which is calculated as $O(K^2 + 2N \cdot T_{ttl})$.

As is shown in Fig. 7, our approach and the genetic algorithm outperformed other beam-hopping algorithms, in our simulations, the execution time per step for MAAC-GS BH and MADDPG PA is measured at 3.2 ms and 2.7 ms, respectively, and the complete two-stage allocation process takes 6.2 ms. In comparison, the genetic algorithm requires 1611.8 ms, which far exceeds the 2 ms decision time slot. Our proposed approach achieving results comparable to the genetic algorithm while keeping execution times within practical limits, making it a viable and efficient solution for real-time decision-making beam hopping systems.

V. CONCLUSION

In this paper, we addressed the resource allocation problem in a multi-beam low Earth orbit satellite communication system by proposing a two-stage cooperative multi-agent reinforcement learning approach. Utilizing full frequency reuse and beam hopping techniques, the system dynamically allocates limited time slot and power resources in response to real-time demand, while accounting for mutual interference among adjacent beam coverage areas. Two dedicated agents were designed to manage the allocation of time slots (determining illuminating areas) and transmit power, respectively. We developed a multi-agent actor-critic model to learn allocation policies and adopted a centralized training with decentralized execution framework, enabling agents to optimize their individual policies by observing others's actions during training, while making independent decisions based on actual traffic requirements. Simulation results demonstrate the effectiveness of the proposed method in mitigating interference, allocating power on-demand, and achieving efficient real-time decision-

making. Future work will focus on three directions: incorporating dynamic channel states into observations for greater adaptability; training on diverse traffic scenarios, including high-load conditions, to improve robustness; and developing a unified framework for the joint optimization of beam selection and power allocation to pursue global solutions. These steps will enhance overall performance in dynamic environments.

VI. ACKNOWLEDGMENT

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REFERENCES

- [1] C.-S. Choi, "Modeling and analysis of downlink communications in a heterogeneous LEO satellite network," *IEEE Trans. Wireless Commun.*, pp. 1–1, Jan. 2024.
- [2] T. Ahmed, A. Alidadi, Z. Zhang, A. U. Chaudhry, and H. Yanikomeroglu, "The digital divide in Canada and the role of LEO satellites in bridging the gap," *IEEE Commun. Mag.*, vol. 60, no. 6, pp. 24–30, June 2022.
- [3] R. S. Hao, J. F. Zhang, S. C. Jin, D. G. Liu, T. J. Li, and Y. J. Cheng, "K-/ka-band shared-aperture phased array with wide bandwidth and wide beam coverage for LEO satellite communication," *IEEE Trans. Antennas Propag.*, vol. 71, no. 1, pp. 672–680, Jan. 2023.
- [4] J. P. Choi, S.-H. Chang, and V. W. Chan, "Cross-layer routing and scheduling for onboard processing satellites with phased array antenna," *IEEE Trans. Wireless Commun.*, vol. 16, no. 1, pp. 180–192, Jan. 2016.
- [5] T. Takahashi, N. Nakamoto, M. Ohtsuka, T. Aoki, Y. Konishi, I. Chiba, and M. Yajima, "On-board calibration methods for mechanical distortions of satellite phased array antennas," *IEEE Trans. Antennas Propag.*, vol. 60, no. 3, pp. 1362–1372, March 2011.
- [6] R. Zhang, J. Ding, Q. Zhou, J. Wang, Y. Gong, and A. Nallanathan, "Deep learning-driven MA-RIS co-design for near-field ISAC: Joint antenna positioning, phase shifting and beamforming optimization," *IEEE Trans. Cogn. Commun. Netw. (Early Access)*, 2025. [Online]. Available: <http://doi.org/10.1109/TCNN.2025.3601852>
- [7] Y. Su, Y. Liu, Y. Zhou, J. Yuan, H. Cao, and J. Shi, "Broadband LEO satellite communications: Architectures and key technologies," *IEEE Wirel. Commun.*, vol. 26, no. 2, pp. 55–61, April 2019.
- [8] P. Gu, R. Li, C. Hua, and R. Tafazolli, "Dynamic cooperative spectrum sharing in a multi-beam LEO-GEO co-existing satellite system," *IEEE Trans. Wireless Commun.*, vol. 21, no. 2, pp. 1170–1182, Feb. 2021.
- [9] X. Mu, Z. Wang, and Y. Liu, "NOMA for integrating sensing and communications toward 6G: A multiple access perspective," *IEEE Wireless Commun.*, vol. 31, no. 3, pp. 316–323, Jan. 2024.
- [10] A. Wang, L. Lei, E. Lagunas, A. I. Pérez-Neira, S. Chatzinotas, and B. Ottersten, "Joint optimization of beam-hopping design and NOMA-assisted transmission for flexible satellite systems," *IEEE Trans. Wireless Commun.*, vol. 21, no. 10, pp. 8846–8858, Oct. 2022.
- [11] Y. Li, Y. Fan, S. Liu, L. Liu, and W. Yang, "Overview of beam hopping algorithms in large scale LEO satellite constellation," in *2021 IEEE 20th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2021, pp. 1345–1351.
- [12] H. Xu, Y. Sun, Y. Zhao, M. Peng, and S. Zhang, "Joint beam scheduling and beamforming design for cooperative positioning in multi-beam LEO satellite networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 4, pp. 5276–5287, April 2024.
- [13] H.-H. Choi, G. Park, K. Heo, and K. Lee, "Joint optimization of beam placement and transmit power for multibeam LEO satellite communication systems," *IEEE Internet Things J.*, vol. 11, no. 8, pp. 14 804–14 813, April 2024.
- [14] S. Yuan, Y. Sun, M. Peng, and R. Yuan, "Joint beam direction control and radio resource allocation in dynamic multi-beam LEO satellite networks," *IEEE Trans. Veh. Technol.*, pp. 1–15, 2024.
- [15] E. Kim, I. P. Roberts, and J. G. Andrews, "Downlink analysis and evaluation of multi-beam LEO satellite communication in shadowed rician channels," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2061–2075, 2024.
- [16] A. I. Aravanis, B. Shankar M. R., P.-D. Arapoglou, G. Danoy, P. G. Cottis, and B. Ottersten, "Power allocation in multibeam satellite systems: A two-stage multi-objective optimization," *IEEE Trans. Wireless Commun.*, vol. 14, no. 6, pp. 3171–3182, June 2015.
- [17] H. Han, X. Zheng, Q. Huang, and Y. Lin, "Qos-equilibrium slot allocation for beam hopping in broadband satellite communication systems," *Wirel. Netw.*, vol. 21, pp. 2617–2630, 2015.
- [18] F. Tian, L. Huang, G. Liang, X. Jiang, S. Sun, and J. Ma, "An efficient resource allocation mechanism for beam-hopping based LEO satellite communication system," in *2019 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, 2019, pp. 1–5.
- [19] N. Gao, Z. Qin, X. Jing, Q. Ni, and S. Jin, "Anti-intelligent UAV jamming strategy via deep Q-networks," *IEEE Trans. Commun.*, vol. 68, no. 1, pp. 569–581, Jan. 2020.
- [20] X. Hu, S. Liu, R. Chen, W. Wang, and C. Wang, "A deep reinforcement learning-based framework for dynamic resource allocation in multibeam satellite systems," *IEEE Commun. Lett.*, vol. 22, no. 8, pp. 1612–1615, Aug. 2018.
- [21] Y. Zhang, X. Hu, R. Chen, Z. Zhang, L. Wang, and W. Wang, "Dynamic beam hopping for DVB-S2X satellite: A multi-objective deep reinforcement learning approach," in *2019 IEEE International Conferences on Ubiquitous Computing and Communications (IUCC) and Data Science and Computational Intelligence (DSCI) and Smart Computing, Networking and Services (SmartCNS)*, 2019, pp. 164–169.
- [22] X. Hu, Y. Zhang, X. Liao, Z. Liu, W. Wang, and F. M. Ghannouchi, "Dynamic beam hopping method based on multi-objective deep reinforcement learning for next generation satellite broadband systems," *IEEE Trans. Broadcast.*, vol. 66, no. 3, pp. 630–646, Sep. 2020.
- [23] G. Xu, F. Tan, Y. Ran, Y. Zhao, and J. Luo, "Joint beam-hopping scheduling and coverage control in multibeam satellite systems," *IEEE Wireless Commun. Lett.*, vol. 12, no. 2, pp. 267–271, Feb. 2023.
- [24] X. Du, X. Hu, Y. Wang, and W. Wang, "Dynamic resource allocation for beam hopping satellites communication system: An exploration," in *2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2022, pp. 1296–1301.
- [25] X. Liao, X. Hu, Z. Liu, S. Ma, L. Xu, X. Li, W. Wang, and F. M. Ghannouchi, "Distributed intelligence: A verification for multi-agent drl-based multibeam satellite resource allocation," *IEEE Commun. Lett.*, vol. 24, no. 12, pp. 2785–2789, Dec. 2020.
- [26] Z. Lin, Z. Ni, L. Kuang, C. Jiang, and Z. Huang, "Dynamic beam pattern and bandwidth allocation based on multi-agent deep reinforcement learning for beam hopping satellite systems," *IEEE Trans. Veh. Technol.*, vol. 71, no. 4, pp. 3917–3930, April 2022.
- [27] R. Lowe, Y. I. Wu, A. Tamar, J. Harb, O. Pieter Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," *Advances in neural information processing systems*, vol. 30, 2017.
- [28] C. R. Tilbury, F. Christianos, and S. V. Albrecht, "Revisiting the gumbel-softmax in maddpg," *arXiv preprint arXiv:2302.11793*, 2023.
- [29] L. Wang, X. Hu, S. Ma, S. Xu, and W. Wang, "Dynamic beam hopping of multi-beam satellite based on genetic algorithm," in *2020 IEEE Intl Conf on Parallel & Distributed Processing with Applications, Big Data & Cloud Computing, Sustainable Computing & Communications, Social Computing & Networking (ISPA/BDCLOUD/SocialCom/SustainCom)*. IEEE, 2020, pp. 1364–1370.
- [30] L. Chen, V. N. Ha, E. Lagunas, L. Wu, S. Chatzinotas, and B. Ottersten, "The next generation of beam hopping satellite systems: Dynamic beam illumination with selective precoding," *IEEE Trans. Wireless Commun.*, vol. 22, no. 4, pp. 2666–2682, April 2022.
- [31] P. Angeletti, D. Fernandez Prim, and R. Rinaldo, "Beam hopping in multi-beam broadband satellite systems: System performance and payload architecture analysis," in *24th aiaa international communications satellite systems conference*, 2006, p. 5376.
- [32] S. Sukhbaatar, R. Fergus *et al.*, "Learning multiagent communication with backpropagation," *Advances in neural information processing systems*, vol. 29, 2016.
- [33] T. Rashid, M. Samvelyan, C. S. De Witt, G. Farquhar, J. Foerster, and S. Whiteson, "Monotonic value function factorisation for deep multi-agent reinforcement learning," *Journal of Machine Learning Research*, vol. 21, no. 178, pp. 1–51, 2020.
- [34] J. Ekelund, R. Vinuesa, Y. V. Khotyaintsev, P. Henri, G. Delzanno, and S. Markidis, "AI in space for scientific missions: Strategies for minimizing neural-network model upload," *2024 IEEE 20th International Conference on e-Science (e-Science)*, pp. 1–10, 2024.
- [35] L. Diana and P. Dini, "Review on hardware devices and software techniques enabling neural network inference onboard satellites," *Remote Sensing*, 2024.